3.1: Polar covalent bonds in water molecules result in hydrogen bonding

Water is so familiar to us that it is easy to overlook its many extraordinary qualities. Following the theme of emergent properties, we can trace water's unique behavior to the structure and interactions of its molecules.

Studied on its own, the water molecule is deceptively simple. It is shaped like a wide V, with its two hydrogen atoms joined to the oxygen atom by single covalent bonds. Oxygen is more electronegative than hydrogen, so the electrons of the covalent bonds spend more time closer to oxygen than to hydrogen; these are ******polar covalent bonds** (see Figure 2.11). This unequal sharing of electrons and water's V-like shape make it a **polar molecule**, meaning that its overall charge is unevenly distributed. In water, the oxygen of the molecule has partial negative charges (δ−), and the hydrogens have partial positive charges (δ+).

The properties of water arise from attractions between oppositely charged atoms of different water molecules: The partially positive hydrogen of one molecule is attracted to the partially negative oxygen of a nearby molecule. The two molecules are thus held together by a hydrogen bond **(Figure 3.2)**. When water is in its liquid form, its hydrogen bonds are very fragile, each only about 1/20 as strong as a covalent bond. The hydrogen bonds form, break, and re-form with great frequency. Each lasts only a few trillionths of a second, but the molecules are constantly forming new hydrogen bonds with a succession of partners. Therefore, at any instant, most of the water molecules are hydrogen-bonded to their neighbors. The extraordinary properties of water emerge from this hydrogen bonding, which organizes water molecules into a higher level of structural order.

Figure 3.2

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    Hydrogen bonds between water molecules.
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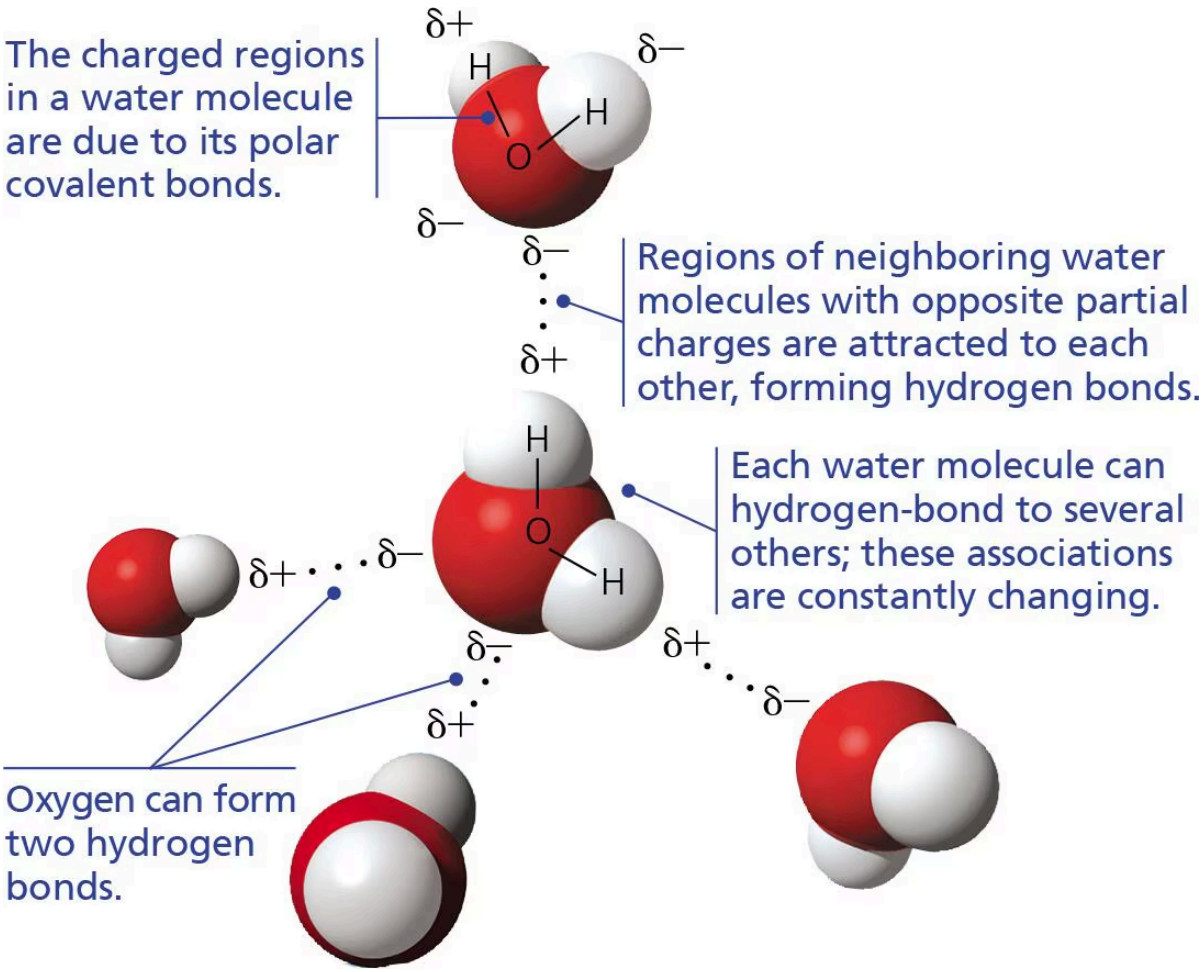


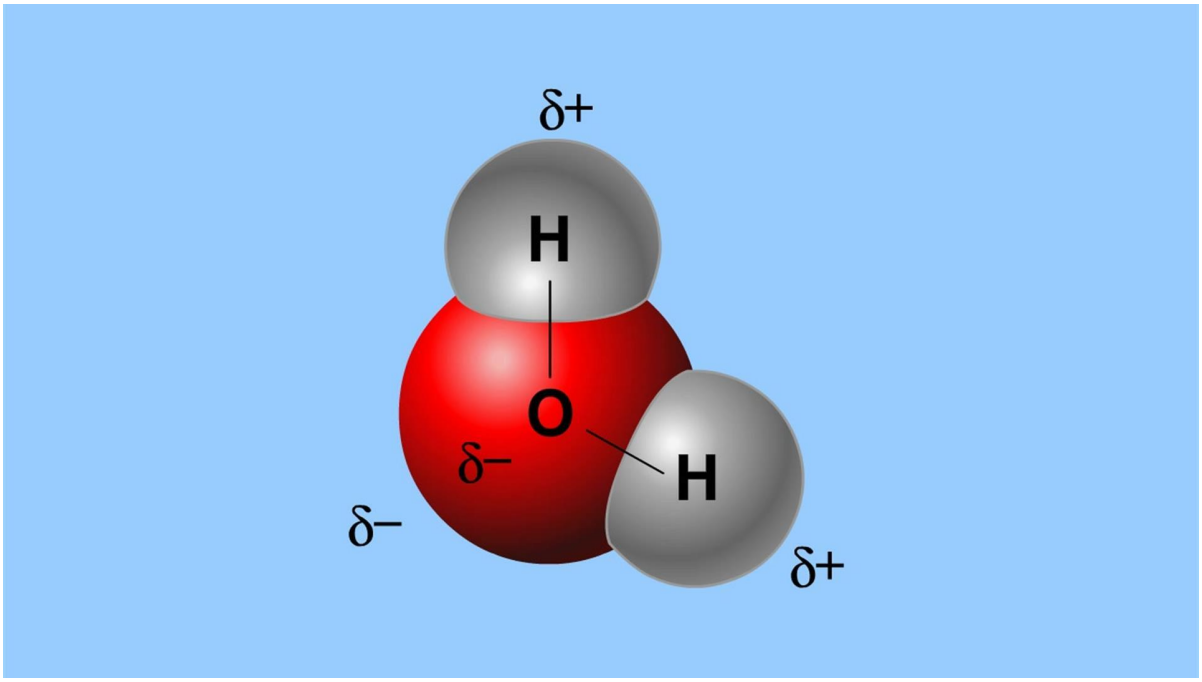
The charged regions in a water molecule are due to its polar covalent bonds. Because of its electron arrangement, oxygen has two regions with partial negative charge. Regions of neighboring water molecules with opposite partial charges are attracted to each other, forming hydrogen bonds. Each water molecule can hydrogen-bond to several others; these associations are constantly changing.

DRAW IT *Draw partial charges on the water molecule at the far left, and draw three more water molecules hydrogen-bonded to it.*

*For suggested answer, see**Appendix A.*

Animation

Polarity of Water



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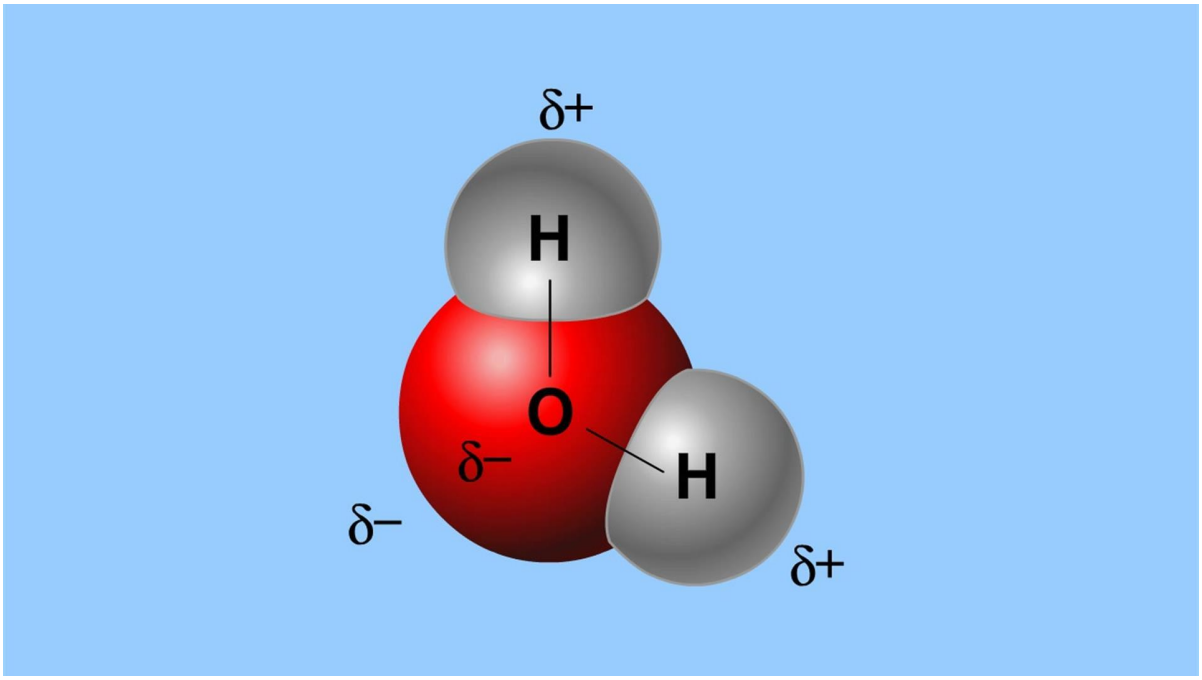

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